Therefore, $a-b=1 \quad \ldots(1)$

$2a+c=5 \quad \ldots(2)$

$2a-b=0 \quad \ldots(3)$

$3c+d=2 \quad \ldots(4)$

$(3) \Rightarrow \quad 2a-b=0$

$2a=b \quad \ldots(5)$

Put $2a=b$ in equation (1), $a-2a=1 \Rightarrow a=-1$

Put $a=-1$ in equation (5), $2(-1)=b \Rightarrow b=-2$

Put $a=-1$ in equation (2), $2(-1)+c=5 \Rightarrow c=7$

Put $c=7$ in equation (4), $3(7)+d=2 \Rightarrow d=-19$

Therefore, $a=-1, b=-2, c=7, d=-19$

Exercise 3.17

1. In the matrix $A=\begin{pmatrix} 8 & 9 & 4 & 3 \\ -1 & \sqrt{7} & \frac{\sqrt{3}}{2} & 5 \\ 1 & 4 & 3 & 0 \\ 6 & 8 & -11 & 1 \end{pmatrix}$, write (i) The number of elements

(ii) The order of the matrix (iii) Write the elements $a_{22}, a_{23}, a_{24}, a_{34}, a_{43}, a_{44}$.

2. If a matrix has 18 elements, what are the possible orders it can have? What if it has 6 elements?

3. Construct a 3×3 matrix whose elements are given by

(i) $a_{ij}=|i-2j|$ (ii) $a_{ij}=\frac{(i+j)^3}{3}$

4. If $A=\begin{pmatrix} 5 & 4 & 3 \\ 1 & -7 & 9 \\ 3 & 8 & 2 \end{pmatrix}$ then find the transpose of A.

5. If $A=\begin{pmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{pmatrix}$ then find the transpose of $-A$.

6. If $A=\begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$ then verify $(A^T)^T=A$

7. Find the values of x, y and z from the following equations

(i) $\begin{pmatrix} 12 & 3 \\ x & 5 \end{pmatrix}=\begin{pmatrix} y & z \\ 3 & 5 \end{pmatrix}$ (ii) $\begin{pmatrix} x+y & 2 \\ 5+z & xy \end{pmatrix}=\begin{pmatrix} 6 & 2 \\ 5 & 8 \end{pmatrix}$ (iii) $\begin{pmatrix} x+y+z \\ x+z \\ y+z \end{pmatrix}=\begin{pmatrix} 9 \\ 5 \\ 7 \end{pmatrix}$

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